A Web-Based Tracking and Inventory Management System for OLV Water Refilling Station

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
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In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **A Web-Based Tracking and Inventory Management System for OLV Water Refilling Station**

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

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Introduction

Project Context

Businesses today are increasingly adopting digital technologies to improve operational efficiency, accuracy, and customer service. According to Laudon and Laudon (2020), information systems play a vital role in helping businesses automate processes, organize records, and provide real-time access to important information. Through the use of web-based systems, businesses are able to reduce manual workloads, minimize errors, and improve the management of daily operations. As technology continues to evolve, small and medium enterprises are also beginning to shift from traditional paper-based methods to computerized systems to remain efficient and competitive. Furthermore, inventory management systems are considered essential tools for businesses that rely on product monitoring and stock control. Stair and Reynolds (2019) explain that computerized inventory systems help organizations maintain accurate records, monitor stock availability, and improve decision-making through real-time updates and reporting features. Web-based systems also provide flexibility and accessibility, allowing users to monitor operations using internet-enabled devices. This makes business operations more organized, reliable, and easier to manage compared to manual methods. In the Philippines, many small businesses are gradually adopting digital systems to improve productivity and service quality. According to the Department of Trade and Industry (DTI) (2021), digital transformation enables small and medium enterprises (SMEs) to streamline business operations, improve data management, and enhance overall efficiency.

However, despite the availability of technology, many small businesses still rely on handwritten records and manual inventory monitoring, which may lead to data loss, inaccurate tracking, delays in retrieving information, and inefficient operations. OLV Water Refilling Station is a family-managed business located at 1095 St. Agatha Road, 5th OLV, Bitano, Legazpi City. The business was originally established as Albay Clear Water Refilling Station in Brgy. 42 Delta Subdivision, Pag-Asa, Rawis, Legazpi City, before it was later renamed to ACW Water Refilling Station and eventually became OLV Water Refilling Station. The business officially started operating on May 1, 2017, and currently refills more or less 100 gallons of water per day for customers within the community. The station provides purified drinking water and related services while handling daily inventory monitoring, customer record management, and operational tracking to ensure continuous service and product availability. Currently, many of the station's processes are still managed manually using logbooks and paper-based records. Although these methods are functional, they can become time-consuming, difficult to organize, and prone to errors as business operations continue to grow. The business experiences challenges in monitoring inventory levels, organizing customer information, updating records, and retrieving important data efficiently. Manual processes may result in inaccurate inventory records, misplaced information, delays in monitoring stock availability, and difficulty in maintaining organized customer records. These issues can affect operational efficiency and may slow down daily business operations. To address these challenges, the proponents propose the development of a Web-based Tracking and Inventory Management System for the OLV Water Refilling Station. The proposed system aims to improve inventory monitoring, customer information management, and

record organization through a centralized web-based platform. The system shall provide a user-friendly interface that enables the admin and staff to monitor inventory, update records, and generate reports efficiently. By implementing the proposed system, OLV Water Refilling Station is expected to improve operational efficiency, reduce manual workload, and enhance the accuracy, accessibility, and organization of business records.

Purpose and Description

The purpose of the project is to develop a Web-Based Tracking and Inventory Management System for OLV Water Refilling Station that aims to improve the management of inventory records and customer information through a centralized digital platform. The project seeks to help the business transition from traditional manual recording methods to a more efficient and organized web-based system. The proposed system will provide the owner and staff with a platform where they can efficiently monitor inventory levels, manage customer records, update information, and generate reports accurately and conveniently. Throughout the development process, the proponents aim to ensure that the system meets the operational needs and requirements of the OLV Water Refilling Station. The proposed system is expected to reduce manual paperwork, minimize errors in inventory tracking, and improve the organization and accessibility of records. By converting manual processes into a web-based platform, the system can help reduce the workload of the staff and improve the efficiency of daily business operations. Furthermore, the system shall provide a user-friendly interface that allows authorized users to monitor inventory movement and customer information in real-time. The implementation of the proposed system is expected to improve operational efficiency,

simplify record management, and support better monitoring and decision-making for the business.

Objectives

The main objective of the study is to develop a Web-Based Tracking and Inventory Management System for OLV Water Refilling Station that aims to improve inventory monitoring, customer record management, and operational efficiency through a centralized web-based platform. The proposed system seeks to help the business transition from manual record-keeping and paper-based inventory monitoring to a more organized and efficient digital system. The system is expected to reduce manual workload, improve the accuracy of inventory records, and provide faster access to customer and inventory information. Furthermore, the proposed system aims to improve data organization and reporting processes to support better monitoring and decision-making for the business.

Specifically, this study aims to:

- Create an Inventory Management Module that allows the admin and staff to record, update, monitor, and manage water gallons, containers, refilling supplies, and other inventory items efficiently.
- Develop an Inventory Tracking Module that enables monitoring of stock availability, inventory movement, gallon/container status, and inventory levels in real-time.

- Establish a Customer Information Module that stores, organizes, and manages customer records, contact information, addresses, and transaction history in a centralized database.
- Create a Report Generation Module that generates inventory reports, customer transaction reports, stock summaries, and operational reports.
- Develop a Scheduling and Monitoring Module that helps staff organize customer orders, delivery schedules, and service requests efficiently.
- Develop a Login and Authentication Module that provides secure system access for authorized users.
- Design an efficient web-based interface that enables users to navigate the system easily and perform operations efficiently.

Scope and Limitations

The scope of the study focuses on improving the inventory monitoring and record management processes of the OLV Water Refilling Station. Currently, the business relies on manual recording methods such as logbooks and paper-based customer records, which may result in delays, inaccurate inventory tracking, misplaced information, and difficulty in monitoring daily operations. As business operations continue to grow, managing inventory and customer information manually becomes time-consuming and less efficient. The proponents aim to address these issues through the development of a Web-Based Tracking and Inventory Management System with the following features.

Scope

The features of the study shall help OLV Water Refilling Station improve inventory monitoring, customer record management, and overall business operations. Through a user-friendly web-based interface, the system shall allow the admin and staff to efficiently record, update, and monitor inventory items such as water gallons, containers, and refilling supplies. The system shall provide real-time inventory tracking to help monitor stock availability, inventory movement, and inventory status, reducing manual errors and improving operational efficiency.

The system shall also include a customer information module that allows users to store, organize, and manage customer records in a centralized database for easier retrieval and monitoring. Authorized users can add, edit, update, and delete inventory and customer information through the CRUD functionality. In addition, the system shall generate reports related to inventory records, customer information, stock summaries, and delivery schedules to support documentation and business monitoring.

For the administrator module, the admin can manage inventory records, customer information, reports, schedules, and user accounts. For the staff module, staff members can monitor inventory levels, update customer information, manage delivery schedules, and generate reports. The system is web-based and can be accessed through internet-enabled devices, providing convenient monitoring and management of daily business operations.

Limitations

The system has several limitations. The proposed system does not include GPS-based real-time delivery tracking due to the additional hardware, internet requirements, and location-based technologies needed for implementation. Online payment integration is not included because the study focuses primarily on inventory management, customer information management, and operational monitoring rather than financial transaction processing.

Advanced accounting, payroll management, and financial reporting features are also excluded from the scope of the study to maintain focus on inventory tracking and customer management functionalities. The proposed system is designed as a web-based application and does not include a dedicated mobile application version. Although the system may provide limited offline access to previously stored information, internet connectivity is still required for data synchronization, report generation, record updates, and access to the complete functionality of the system.

The proposed system does not include SMS notification services, integrated chat or messaging features, and GCash payment processing. Although these features were identified during the interview as potential enhancements, they are excluded from the current version of the system to maintain focus on inventory management, customer information management, scheduling, and operational monitoring. These features may be considered for future system enhancements.

Review of Related Literature/Studies/Systems

This chapter presents the related literature and studies reviewed by the proponents to gather relevant information and strengthen the foundation of the proposed study. The related literature and studies provide concepts, ideas, and findings from both local and foreign sources that are connected to the development of the proposed Web-Based Tracking and Inventory Management System for OLV Water Refilling Station. These references help the proponents understand the importance of inventory management, customer record management, web-based systems, and business automation in improving operational efficiency and accuracy.

Review of Related Literature

Foreign Studies

Turban, E., Pollard, C., and Wood, G. (2021) – Information Technology for Management

According to Turban, E., Pollard, C., and Wood, G. (2021), technology significantly improves business management by automating processes, reducing repetitive tasks, and increasing workflow efficiency. They also explained that integrating business functions such as inventory and customer management into a single platform enhances decision-making and reporting accuracy.This literature is related to the proposed system because it supports the integration of inventory tracking and customer management into a unified and centralized system for improved business operations.

Laudon, K. C., and Laudon, J. P. (2022) – Management Information Systems: Managing the Digital Firm

According to Laudon and Laudon (2022), Management Information Systems (MIS) play a vital role in improving organizational efficiency through automation, organized data management, and real-time information access. The study explained that computerized systems reduce manual workload, minimize operational errors, and improve the accuracy of business records. It also emphasized that inventory systems help businesses maintain accurate stock monitoring while improving customer service and operational productivity. This literature supports the proposed study because the proposed system focuses on automating inventory monitoring and customer record management to improve efficiency and accuracy in business operations.

Stair, R. M., and Reynolds, G. W. (2021) – *Principles of Information Systems*

According to Stair and Reynolds (2021), information systems enable organizations to improve efficiency, enhance decision-making, and support business operations through the effective use of technology. The authors explained that web-based systems provide users with convenient access to information and services while allowing organizations to centralize and manage data more efficiently. They also emphasized that integrated information systems improve communication, data processing, and overall organizational productivity. This literature is related to the proposed system because it supports the implementation of a web-based platform that centralizes inventory records and customer information to improve operational efficiency and data management.

Kendall, K. E., & Kendall, J. E. (2024) – *Systems Analysis and Design*

According to Kendall and Kendall (2024), systems analysis and design play a vital role in developing information systems that effectively address organizational needs. The authors explained that properly designed systems improve data processing, enhance communication, and provide better support for decision-making. They also emphasized that web-based systems enable organizations to centralize information and improve accessibility, leading to increased operational efficiency. This literature is related to the proposed system because it supports the development of a web-based platform that integrates inventory management and customer records to improve business operations and information accessibility.

Baltzan, P. (2021) – *Business Driven Information Systems*

According to Baltzan (2021), information systems help organizations improve efficiency, support decision-making, and enhance communication through integrated technologies. The author explained that businesses use information systems to automate processes, centralize data, and improve operational performance. The book also emphasized that web-based systems enable organizations to access and manage information efficiently, leading to better productivity and customer service. This literature is related to the proposed system because it supports the implementation of a web-based platform that centralizes inventory records and customer information to improve operational efficiency and data management.

Local Literature

Department of Trade and Industry (DTI) (2021) – Digital Transformation of MSMEs

According to the Department of Trade and Industry (DTI) (2021), digital transformation enables micro, small, and medium enterprises (MSMEs) to improve business operations through the adoption of digital technologies and information systems. The agency emphasized that digital tools help businesses improve efficiency, manage records more effectively, and enhance customer service. It also highlighted that technology adoption allows businesses to remain competitive in a rapidly changing market environment. This literature supports the proposed study because the system aims to help OLV Water Refilling Station transition from manual record-keeping to a digital platform for better inventory and customer management.

Department of Information and Communications Technology (DICT) (2022) – Digital Transformation Program

According to the Department of Information and Communications Technology (DICT) (2022), information and communication technologies play a significant role in improving productivity, operational efficiency, and service delivery among organizations and businesses. The agency explained that digital systems help streamline business processes, improve accessibility to information, and support data-driven decision-making. This literature is related to the proposed study because it supports the implementation of a web-based system that provides efficient access to inventory and customer information.

Philippine Statistics Authority (PSA) (2023) – Information and Communications Technology Statistics

According to the Philippine Statistics Authority (PSA) (2023), the use of information and communications technology among businesses continues to increase as organizations recognize the benefits of digital solutions in managing operations and improving efficiency. The report highlighted that businesses using computerized systems experience improved data management, faster information retrieval, and better operational performance. This literature supports the proposed system because it demonstrates the growing importance of digital technologies in improving business processes and record management.

National Economic and Development Authority (NEDA) (2023) – Philippine Development Plan 2023–2028

According to the National Economic and Development Authority (NEDA) (2023), digitalization is a key strategy for improving productivity, innovation, and business competitiveness in the Philippines. The plan emphasizes the importance of adopting digital technologies to modernize operations, improve service delivery, and strengthen data management capabilities. This literature supports the proposed study because the development of a web-based inventory and tracking system contributes to the country's goal of promoting digital transformation among businesses.

Department of Science and Technology (DOST) (2022) – Digital Transformation Initiatives for Small Businesses

According to the Department of Science and Technology (DOST) (2022), digital systems help organizations improve operational efficiency, automate routine tasks, and enhance information management. The agency highlighted that technology-driven solutions reduce manual processes and support better monitoring of business activities through real-time information access. This literature is related to the proposed study because the proposed system seeks to automate inventory monitoring and customer record management through a centralized web-based platform.

Review of Related Studies

Foreign Studies

Erameh, K. B., and Odoh, B. I. (2021) – Design and Implementation of a Web-Based Inventory Control System Using a Small Medium Enterprise (SME) as a Case Study

According to Erameh and Odoh (2021), web-based inventory control systems improve the management of stock records by automating inventory monitoring and reducing manual errors. Their study found that businesses using computerized inventory systems experienced better accuracy in stock tracking, improved record management, and faster retrieval of information. The researchers emphasized that web-based solutions help organizations manage inventory efficiently while supporting decision-making through real-time data access. This study supports the proposed system because it demonstrates

how a web-based inventory management platform can improve inventory monitoring, record accuracy, and operational efficiency.

Misahuaman, G., Daza, A., and Zavaleta, E. (2021) – Web-Based Systems for Inventory Control in Organizations: A Systematic Review

According to Misahuaman, Daza, and Zavaleta (2021), web-based inventory systems play a significant role in improving inventory control and organizational efficiency. Their review revealed that inventory management platforms provide accurate stock monitoring, minimize inventory losses, and improve accessibility to information through centralized databases. The study also highlighted the importance of digital technologies in supporting efficient inventory operations. This study is related to the proposed system because it supports the implementation of a web-based inventory tracking platform that enables accurate inventory monitoring and information management.

Agboola, F. F., Malgwi, Y. M., Mahmud, M. A., & Oguntoye, J. P. (2022) – Development of a Web-Based Platform for Automating an Inventory Management of a Small and Medium Enterprise

According to Agboola et al. (2022), manual inventory management often leads to inaccurate records, delayed updates, and difficulty in monitoring stock levels. The researchers developed a web-based inventory management platform that automated inventory recording, stock monitoring, and report generation. The study found that the system improved inventory accuracy, reduced manual errors, and enhanced operational efficiency. This study supports the proposed system because it demonstrates how a

web-based inventory management system can improve stock monitoring and record management for small businesses.

Teerasoponpong, S., & Sopadang, A. (2022) – Decision Support System for Adaptive Sourcing and Inventory Management in Small and Medium-Sized Enterprises

According to Teerasoponpong and Sopadang (2022), inventory management systems help SMEs improve stock control, decision-making, and operational efficiency through data-driven monitoring and inventory management strategies. The study highlighted that computerized inventory systems enable businesses to reduce stock shortages and improve resource utilization. This study supports the proposed system because it emphasizes the importance of inventory monitoring and computerized management systems in improving business operations.

Alshourah, S., Alshurideh, M., and Al Kurdi, B. (2023) – The Impact of Digital Inventory Management Systems on Business Performance

According to Alshourah, Alshurideh, and Al Kurdi (2023), digital inventory management systems significantly improve inventory accuracy, operational efficiency, and decision-making within organizations. Their study found that businesses utilizing digital inventory solutions experience reduced stock discrepancies, better inventory visibility, and more effective resource management. The researchers emphasized that inventory automation contributes to improved business performance by providing real-time access to inventory data and reducing manual processes. This study supports the proposed

system because it demonstrates how a digital inventory management platform can improve inventory monitoring, record accuracy, and operational efficiency for businesses such as OLV Water Refilling Station.

Local

Bautista, J. V., Jr., & Young, M. N. (2022) – Effective Inventory Management System in Efficient Supply and Distribution Management in One Manufacturer of Food Seasoning Products in the Philippines

According to Bautista and Young (2022), inventory management systems play an essential role in ensuring efficient supply and distribution processes. Their study emphasized that accurate inventory monitoring prevents stock shortages and improves the ability of organizations to meet customer demand. The researchers found that effective inventory management contributes to improved operational performance and better business productivity. This study supports the proposed system because it highlights the importance of maintaining accurate inventory records and improving inventory monitoring to enhance operational efficiency.

De Los Santos, J. R. N., Suyom, J. R., Comora, M. B., & De Los Santos, G. (2021) – Imbentaryo App: An Intelligent Inventory and Decision Support System

According to De Los Santos et al. (2021), integrating forecasting algorithms and SMS technology into inventory systems improves planning, monitoring, and inventory control. Their study revealed that the system enhanced decision-making and provided efficient management of supplies and equipment. The researchers concluded that intelligent

inventory systems contribute to better operational effectiveness and productivity. This study supports the proposed system because it demonstrates how digital inventory systems improve monitoring and decision-making processes.

Ahmad, A. (2022) – e-AIMSS (Electronic Asset Inventory and Management System in Schools) for Resource Optimization and Organizational Productivity

According to Ahmad (2022), digital inventory systems improve asset monitoring and optimize organizational productivity through accurate and centralized record management. The study showed that electronic inventory systems contribute to efficiency, reliability, and sustainability in managing organizational resources. This study is related to the proposed system because it demonstrates the effectiveness of digital inventory systems in improving resource monitoring and operational efficiency.

Parilla, E., Evangelista, J., Aurelio, R., & Bullalayao, C. (2022) – Inventory Management Practices and Service Delivery of Healthcare Facilities in Ilocos Norte, Philippines

According to Parilla et al. (2022), effective inventory management practices positively influence service delivery and organizational performance. Their findings indicated that maintaining accurate inventory records contributes to improved efficiency and customer satisfaction. This study supports the proposed system because it emphasizes the importance of inventory management in improving organizational effectiveness and service quality.

Ines, M. M. (2025) – Exploring Financial Management and Performance of Small and Medium Enterprises

According to Ines (2025), effective financial management practices such as budgeting, record keeping, reporting, and inventory management contribute significantly to the performance and sustainability of small and medium enterprises (SMEs). The study found that businesses that maintain organized records and proper inventory management practices are more capable of improving operational efficiency, supporting decision-making, and achieving better business performance. It also emphasized that technology adoption can enhance financial transparency, inventory monitoring, and overall business management. This study supports the proposed Web-Based Tracking and Inventory Management System because it highlights the importance of accurate inventory management, organized record keeping, and technology adoption in improving business operations. The findings reinforce the need for a centralized digital system that can help OLV Water Refilling Station manage inventory records, customer information, and operational data more efficiently.

Synthesis

Based on the reviewed literature and studies, it is evident that web-based systems, inventory management systems, and business automation technologies play a crucial role in enhancing operational efficiency, improving inventory monitoring, and facilitating customer record management. Both local and foreign studies emphasized that computerized systems reduce manual workload, minimize errors, improve data organization, and provide real-time access to information. The reviewed literature also

highlighted the importance of web-based platforms in improving accessibility and flexibility for businesses. Inventory management systems have been proven effective in maintaining accurate stock records and improving monitoring processes, while customer management systems have enhanced data organisation and retrieval efficiency. However, most existing studies focus mainly on general inventory systems, POS systems, or customer management systems separately. There are limited studies specifically designed for water refilling stations that combine inventory tracking and customer information management into one centralized web-based platform. Therefore, the proponents aim to develop a Web-Based Tracking and Inventory Management System for OLV Water Refilling Station that integrates inventory monitoring, customer information management, record updating, and report generation into a single system. The proposed system is expected to improve operational efficiency, reduce manual processes, enhance record accuracy, and provide a more organized way of managing daily business operations.

Methodology

This chapter presents the methods and procedures used to complete the development of the proposed Web-Based Tracking and Inventory Management System for OLV Water Refilling Station. The study employed the Agile methodology, which emphasizes iterative development, collaboration, and the ability to adapt to changes throughout the project lifecycle. By using an Agile approach, the proponents can ensure that the system's modules, such as inventory tracking and customer management, are continuously refined based on functional requirements and testing outcomes.

Figure 1.0: Agile Development Model



Figure 1.0 illustrates the Agile Development Model used in the development of the proposed Web-Based Tracking and Inventory Management System for OLV Water Refilling Station. Agile is a software development methodology that focuses on continuous improvement, collaboration, flexibility, and iterative development. Unlike traditional development approaches, Agile divides the project into smaller and manageable phases called iterations, allowing the proponents to continuously evaluate,

improve, and refine the system throughout the development process. The Agile Development Model is suitable for the proposed system because it allows the proponents to gradually develop the inventory management, customer information management, scheduling, and reporting modules while continuously incorporating feedback from the client. Through repeated testing and improvements, the system can better meet the operational requirements of OLV Water Refilling Station.

Requirements Phase

The Requirements Phase involves gathering and analyzing the needs of OLV Water Refilling Station. During this phase, the proponents conducted interviews and observations with the business owner to identify the challenges encountered in the current manual inventory monitoring and customer record management processes. The information gathered helped determine the essential features of the proposed system, including inventory management, inventory tracking, customer information management, scheduling, report generation, and user authentication. The identified requirements served as the foundation for the development of the system.

Design Phase

During the Design Phase, the proponents prepared the overall structure and layout of the system. This phase included the creation of database designs, Data Flow Diagrams (DFD), user interface prototypes, process flows, and system architecture. The design Seasily access and manage inventory records, customer information, delivery schedules, and reports. Proper planning during this phase ensured that the system would be organized, efficient, and aligned with the operational needs of the business.

Build Phase

In the Build Phase, the proponents developed the actual system based on the approved requirements and designs. The system modules were coded and integrated using the selected technologies. The development included the Inventory Management Module, Inventory Tracking Module, Customer Information Module, Scheduling and Monitoring Module, Report Generation Module, and Login and Authentication Module. Each module was developed incrementally to ensure proper functionality and system integration.

Test Phase

The Test Phase focused on evaluating the functionality, performance, and reliability of the developed system. The proponents conducted testing to identify errors, bugs, and inconsistencies that could affect system operations. The testing process verified whether the inventory records, customer information, scheduling functions, and report generation features performed according to the specified requirements. Any issues identified during testing were documented for correction and improvement.

Change Phase

After testing, the Change Phase involved applying modifications and improvements based on the results of system evaluation and user feedback. Necessary adjustments were made to enhance system usability, improve functionality, correct identified errors, and ensure that the system effectively addressed the operational requirements of OLV Water Refilling Station. This phase demonstrates the flexibility of Agile development in responding to changes and improvements throughout the project.

Next Iteration Phase

The Next Iteration Phase allows the development cycle to continue by repeating the previous phases whenever improvements or additional features are needed. Through continuous iterations, the proponents can further enhance the system's functionality, performance, and user experience. This iterative approach ensures that the system continuously evolves to meet changing business needs and operational requirements.

System Testing Phase

Once all iterations have been completed, the system undergoes comprehensive System Testing. This phase ensures that all modules work together properly and that the entire system performs according to its intended functions.

The testing verifies the accuracy of inventory monitoring, customer information management, scheduling functions, report generation, and user authentication. This phase helps ensure that the system is stable, secure, and ready for deployment.

Release Phase

The Release Phase is the final stage of the Agile Development Model. During this phase, the completed Web-Based Tracking and Inventory Management System is deployed and implemented for actual use at OLV Water Refilling Station. The admin and staff can now utilize the system for managing inventory records, monitoring stock availability, organizing customer information, scheduling deliveries, and generating reports. The deployment of the system is expected to improve operational efficiency, reduce manual workload, and enhance the accuracy and accessibility of business records.

Technical Background

- **Technologies to be used**

Frontend

The frontend focuses on providing an intuitive and responsive user experience, allowing the staff and admin to navigate the system across various devices.

- **HTML(HyperText Markup Language):** Used as the structural backbone of the web application. It defines the organization of the dashboard, inventory forms, and customer data tables.
- **CSS (Cascading Style Sheets):** Employed to manage the visual aesthetics, ensuring that the interface is clean and professional.
- **Bootstrap:** Bootstrap is a front-end framework used to accelerate the development process and create a responsive web interface. It provides pre-designed components, grid systems, and styling utilities that ensure compatibility across different screen sizes and devices. By using Bootstrap, the system achieves a modern appearance while reducing development time and improving usability.

Backend Technologies

The backend architecture is designed to handle business logic, secure user authentication, and seamless data flow between the user interface and the database.

Node.js

Node.js is a JavaScript runtime environment used to execute server-side code. It enables the system to process requests, manage data, and support system functionality efficiently. But it also helps to handle the business logic, user authentication, database communication, and request processing for the system.

Database Management

To ensure data integrity and fast retrieval of historical records, a relational database management system (RDBMS) is implemented.

SQLite

SQLite is a lightweight relational database management system used to store and manage inventory records, customer information, user accounts, and system-generated reports. It is easy to set up, requires minimal configuration, and is suitable for small to medium-sized business applications.

Framework / Styling Tool

Tailwind CSS

Tailwind CSS was used as a utility-first CSS framework to make styling faster and more consistent. It helped the developers design responsive pages without writing too much custom CSS.

Table 1.0 Calendar of Activities

Legends:

Ongoing –

Resource

Table 2.0 Hardware Requirements (Desktop)

Components	Description / Specification	Recommended Hardware
RAM	4 GB or higher	8 GB or higher
Storage	4 GB	8 GB or higher
CPU	1.4 GHz	1.6 GHz or higher

Table 2.0 presents the hardware requirements needed to run the proposed system efficiently. The minimum RAM requirement is 4 GB, while 8 GB or higher is recommended for better performance. For storage, at least 4 GB of available space is required, while 8 GB or higher is recommended. The minimum processor speed requirement is 1.4 GHz, although a processor with a speed of 1.6 GHz or higher is recommended to ensure smoother system operation.

Table 3.0 Software Requirements for Development and Testing

Software Category	Tools Technologies
Operating System	Windows 10 or Window 11
Code Editor	Visual Studio Code
Backed Development	Node.js
Database Management	SQLite
Frontend Development	HTML, CSS, Bootstrap
UI Framework	Tailwind CSS
Version Control	Git, GitHub
Web Browsers	Google Chrome, Microsoft Edge
Documentation Tools	Microsoft Word, Google Docs, WSP Office
Diagramming Tools	Draw.io

The software requirements for developing and testing the proposed system include Windows 10 or Windows 11 as the operating system and Visual Studio Code as the code editor. Node.js is used for backend development, while SQLite manages the database. HTML, CSS, Bootstrap are utilized for frontend development, together with Tailwind CSS for responsive interface design. Git and GitHub are used for version control and project collaboration. Google Chrome and Microsoft Edge are used for browser testing, while Microsoft Word, Google Docs, WPS, and lastly Draw.io are utilized for documentation and diagram creation.

Requirements Analysis

Introduction

The study employs a descriptive-developmental research design. The descriptive method is used to analyze the current operational challenges encountered by the OLV Water Refilling Station. In contrast, the developmental method focuses on designing and developing the proposed system solution.

The proponents gather data through interviews, observation, and questionnaires conducted with the business owner and staff members. These methods help identify the current problems in inventory monitoring, customer management, and record organization. System testing is also conducted to evaluate the functionality, usability, reliability, and effectiveness of the developed system.

Experimental

The experimental section of the study demonstrates the transition from the current manual operations of the OLV Water Refilling Station to the proposed digital system.

This process involves replacing traditional paper-based logbooks and manual customer records with a centralized web-based platform designed to improve inventory tracking and operational efficiency. The experiment focuses on implementing automated modules for real-time stock monitoring, organized customer data management, and accurate report generation to eliminate the delays and errors associated with the current manual condition. By converting these manual workflows into a synchronized system accessible via internet-enabled devices, the study aims to enhance the overall accuracy and accessibility of business records for better operational management.

Result and Analysis

The data gathered through interviews and direct observation revealed that OLV Water Refilling Station currently relies on manual record-keeping using logbooks and notebooks for inventory monitoring, customer information management, order processing, and delivery scheduling. Although these methods are functional, they become increasingly difficult to manage as the number of customers, deliveries, and inventory items increases.

Based on the interview with the business owner, several operational challenges were identified. These include difficulties in tracking the status of water gallons and containers, scheduling deliveries efficiently, monitoring customer information, and maintaining updated customer records. The owner also reported instances of late deliveries, duplicate or confusing orders, and challenges in contacting customers when their contact information changes.

The observation conducted by the proponents further confirmed that inventory monitoring, customer record management, scheduling, and report preparation are performed manually. These processes require additional time and effort and may affect the efficiency of daily business operations. The findings indicate the need for a computerized system capable of centralizing inventory and customer information, improving order and delivery monitoring, and generating organized reports.

Therefore, the proposed Web-Based Tracking and Inventory Management System is expected to address these issues by providing real-time inventory monitoring, customer information management, delivery scheduling, and automated report generation. The implementation of the system is expected to improve operational efficiency, reduce manual workload, and enhance the accuracy and accessibility of business records.

Interview and Observation

Interview

The proponents conducted an interview with the owner of OLV Water Refilling Station to gather information regarding the current inventory monitoring and customer record management processes of the business. The interview focused on identifying the operational challenges encountered in manual recording, inventory tracking, customer information management, and report organization. Based on the interview, the owner stated that the current process, which relies on logbooks and paper-based records, is time-consuming and prone to errors. Monitoring inventory levels and organizing customer records become increasingly difficult as the volume of transactions grows. Retrieving records and generating reports manually may also cause delays and

inefficiencies in daily operations. The owner emphasized the need for a computerized system that can simplify inventory monitoring, improve record organization, and generate accurate reports to support business operations.

Observation

The proponents also conducted direct observation of the daily operations of OLV Water Refilling Station. During the observation, it was identified that inventory monitoring, customer record management, and report preparation were performed manually. The manual process required additional time and effort in recording, updating, and retrieving information, which may affect operational efficiency and productivity. The observation further revealed that the business would benefit from a web-based system capable of automating inventory tracking, customer information management, scheduling, and report generation. These findings support the need for the proposed system to improve the overall management of business operations.

Source of Data

This study utilizes data gathered through interviews and direct observation conducted at the OLV Water Refilling Station. The owner provided detailed information regarding the current operational processes, inventory monitoring practices, and challenges encountered in manual record management. Additional data were gathered through observation of the daily business operations to identify the system requirements needed for the proposed Web-Based Tracking and Inventory Management System.

Requirements Documentation

The Web-Based Tracking and Inventory Management System serves as an agreement between the business owner and the proponents regarding the features, functions, and objectives. It defines the system's purpose, operational requirements, and expected functionalities needed to improve inventory monitoring, customer record management, and overall business operations. This ensures that the development process addresses the current challenges encountered by the business and supports the transition from manual record-keeping to a more organized and efficient web-based platform. The system's major features include inventory management and tracking, customer information management, record updating, report generation, and secure login and authentication for authorized users, such as the admin and staff. It also serves as a guide throughout the system development process to ensure that the proposed system meets the operational needs of the OLV Water Refilling Station. Furthermore, the system establishes functional requirements for real-time inventory monitoring, centralized customer record storage, accurate report generation, and secure data management to improve operational efficiency, reduce manual errors, and enhance the organization and accessibility of business records.

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APPENDICES

APPENDIX A. RESOURCE PERSONS

Renato P. Lim

Business Owner of OLV Water Refilling Station

Paul Danielle Ponce Yanzon

STI College Legazpi Capstone 1 Project Adviser

Ms. Maria Diejhey B. Martinez, MIT

STI College Legazpi Lead Panelist

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

Juan Rafael M. Gomez



Tagas, Daraga, Albay

raffytaffygomez12@gmail.com

09662881528

Educational Background

Senior High School	2021-2023	Divine Word College of Legazpi
Junior High School	2017-2021	Philippine Women's University
Level	**Dates**	**Name**
Elementary	2010-2017	Ellinwood Christian School
Tertiary	2023-Present	STI College Legazpi

Professional or Volunteer Experience

Dates	**Experience/Job Title**	**Name and Address of Company or Organization**

Affiliations

Dates	**Name of Organization**	**Position**

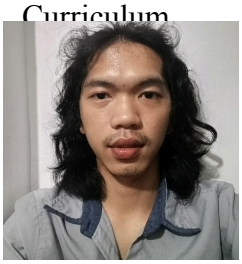
Skills

Skills	**Level of Competency**	**Date Acquired**

Training, Seminars, or Workshops Attended

Curriculum

Vitae of



Jim Kirby Echague

Poblacion, Rapu-Rapu, Albay

Jimkirbygiducos@gmail.com

(+63)96569259114

Educational Background

Level	Dates	Name
Tertiary	2023-Present	STI College School
Senior High School	2019-2021	Rapu-Rapu National HighSchool
Junior High School	2015-2019	Bicol College Junior Highschool
Elementary	2009-2015	Rapu-Rapu Elementary Central School

Professional or Volunteer Experience

Dates	Experience/Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

Affiliations

Dates	Name of Organization	Position
N/A	N/A	N/A

Skills

Skills	Level of Competency	Date Acquired

Training, Seminars, or Workshops Attended

N/A	**N/A**

Curriculum Vitae of



John Manro B. Clemente

Tabaco, Legazpi City, Albay

manro.clementee@gmail.com

(+63)9945010236

Educational Background

Level	Dates	Name
Tertiary	2023-Present	STI College School
Senior High School	2017-2019	AMA Legazpi City
Junior High School	2012-2016	Reyes Computer Oriented School
Elementary	2005-2011	Reyes Computer Oriented School

Professional or Volunteer Experience

Dates	Experience/Job Title	Name and Address of Company or Organization
N/A	**N/A**	**N/A**

Affiliations

Dates	Name of Organization	Position
N/A	**N/A**	**N/A**

Skills

Skills	Level of Competency	Date Acquired
HTML	Beginner	2023
Java	Beginner	2023
IT support	Intermediate	2012

Training, Seminars, or Workshops Attended

N/A	**N/A**

Curriculum Vitae of

Reinalyn R. lim

Bitano, Legazpi City, Albay

reinalynlim219@gmail.com

09199652671



Educational Background

Level	Dates	Name
Tertiary	2023-Present	STI College School
High School	2021-2023	Bicol College Senior High School
Junior High School	2017-2021	Bicol College Junior HighSchool
Elementary	2011-2017	Bicol College Elementary School

Professional or Volunteer Experience

Dates	Experience/Job Title	Name and Address of Company or Organization
N/A	**N/A**	**N/A**

Affiliations

Dates	Name of Organization	Position
N/A	**N/A**	**N/A**

Skills

Skills	Level of Competency	Date Acquired

Training, Seminars, or Workshops Attended

N/A	**N/A**

APPENDIX C. EVALUATION TOOL/TEST/DOCUMENTS

Part 1: Business Profile

-What is the name of your water filling station?

OLV refilling station
formerly St Agatha DD. 5th OLV Bitano Leg. Cath

-How long has your business been in operation?

10 years

-How many employees do you currently have?

2

-What products and services do you offer?

purified water
pet bottle ~~refation~~ ocarsiu / orders

-What are your hours of operation?

8 hrs

Delivery vehicles

Chariot - 1
kabal | kabal - 2

payments method

1 CASH
2 monthly cheque
3 Daily

Part 2: Current Operations

-How do you currently handle customer orders? (walk-in, delivery, text, call)

mostly text | call | messenger

-How do you record customer orders and sales?

log book

-How do you monitor inventory (water containers, supplies)?

Increase of water bills sales big
Decrease of water bills sales low

-How do you manage delivery schedules?

list of customers and schedules

-How do you track customer information?

to track true cp. message/call

-How much is your water per gallon?

35
60

-How many water products do you sell in a typical day?"

100 gallons

Part 3: Problems and Challenges

-What problems do you experience in managing orders and deliveries?

late order

-Do you experience difficulty in tracking inventory? gallons

mostly Container tracking / Schedule pick up fints gallons before ~~delo~~ deliver.

-Are there delays or errors in recording orders?

no

-Is it difficult to monitor daily operations? Why?

None

-What challenges do you face in managing customer records?

Changes Cp. No. Account PB.

Part 4: System Improvements

-What improvements would you like to see in your current system to make operations easier and more efficient?

Tracking orders / earlier ang orders Customers

-Would a system that tracks orders and deliveries in real time help? Why?

-How important is inventory tracking to your daily operations?

Very important to monitor the Supplies Just Seal of the gallons.

-Would automated notifications (SMS) for customers improve your service?

Yes Important.

-What features would you like in a web-based management system?

~~To~~ page

-Do you think having a dashboard summary (sales, orders, status of inventory)? Why?

Helpful / Malakatulong yan

Part 5: System Adoption

-Are you ready to use a web-based system?

Yes

-Do you think it will be easy for your staff to learn?

Yes

-What are your concerns about using a system?

maybe right / online / offline system.

-What would make the system user-friendly for you?

Yes dapat Users friendly your System not complecated.

Part 6: Additional Insights

-What reports do you need to monitor your business operations, such as inventory, delivery, and customer records?

Monitor: container / sales Kang Creadit. / utang.
Delivery / gas - receipt.

-What common complaints do customers have?

late dilivery / why late delelivery. / late message.

-How can technology improve your service?

good / accurate and credible.

-What improvements or changes do you want to implement to make your operations faster and more efficient?

it depend on your System.

occasional wedding / baptism
Canendeia.

* dispenser rent. / service with quota)

30 gallons a lead / monthly.

* Containers for sale with purified water

₱ 20 — monthly customer orders for additional

APPENDIX D. Data Flow Diagram

Data Flow Diagram (DFD)

The Data Flow Diagrams (DFDs) of the existing and proposed systems for the OLV Water Refilling Station and Web-Based Inventory Management System illustrate the movement of information within each system. These diagrams highlight the processes, data flows, and external entities involved in both setups. They emphasize how the proposed system improves data flow efficiency, enhances accuracy, and supports better overall operational performance for the refilling station.

Figure 2.0 Existing Context Diagram

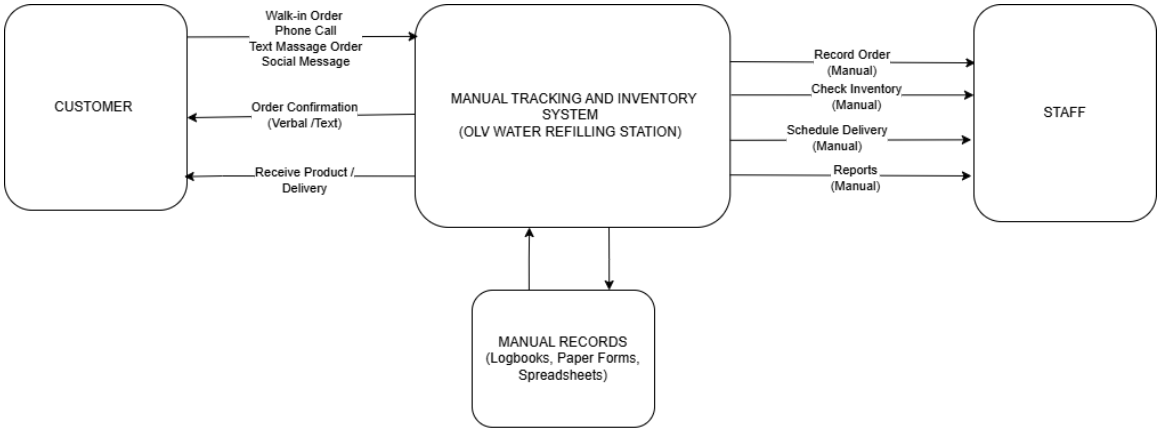


Figure 2.0 - This diagram illustrates the overall flow of information in the current manual system used by OLV Water Refilling Station. It shows how customers and staff interact with the system through walk-in transactions, phone calls, and text message orders. The system relies on manual processes for recording and managing information.

Figure 2.1 Existing Level 1 Data Flow Diagram

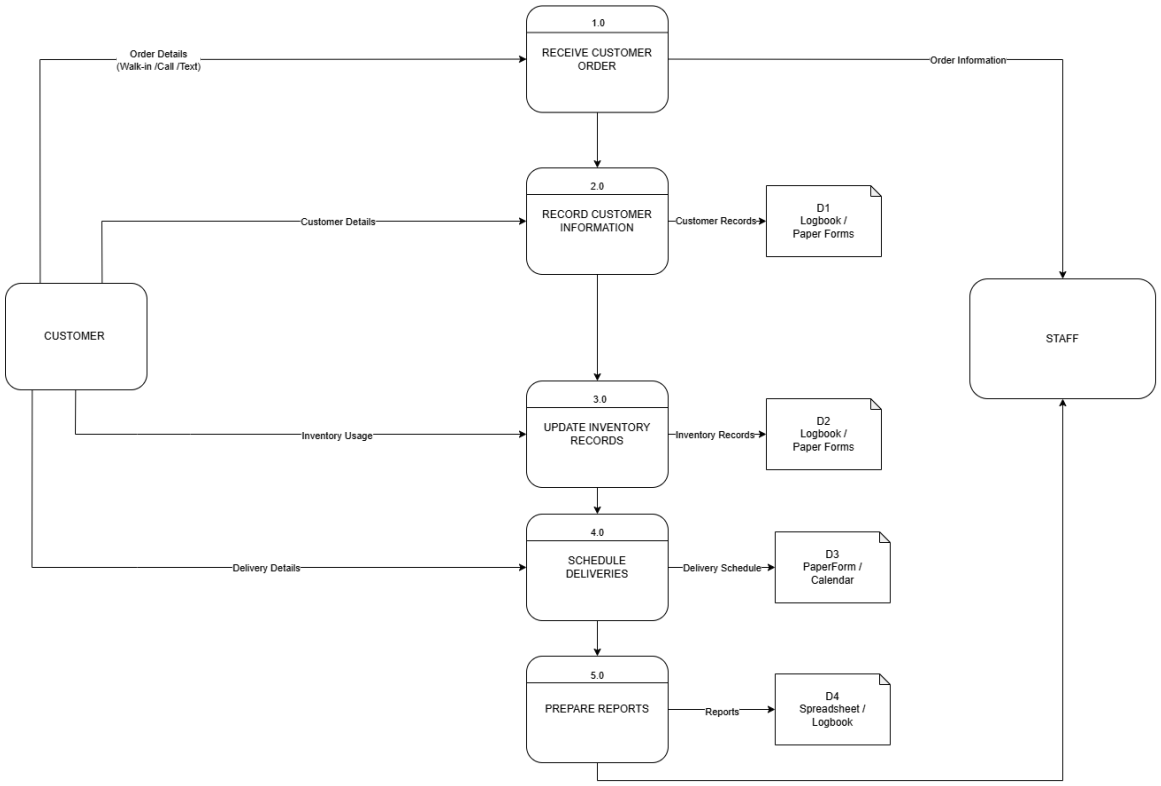


Figure 2.1 - This diagram presents the major processes in the existing manual system, including receiving customer orders, recording customer information, updating inventory records, scheduling deliveries, and preparing reports. These processes are carried out using logbooks, paper forms, and spreadsheet files.

Figure 2.2 Existing Level 2 Data Flow Diagram – Inventory Recording Process

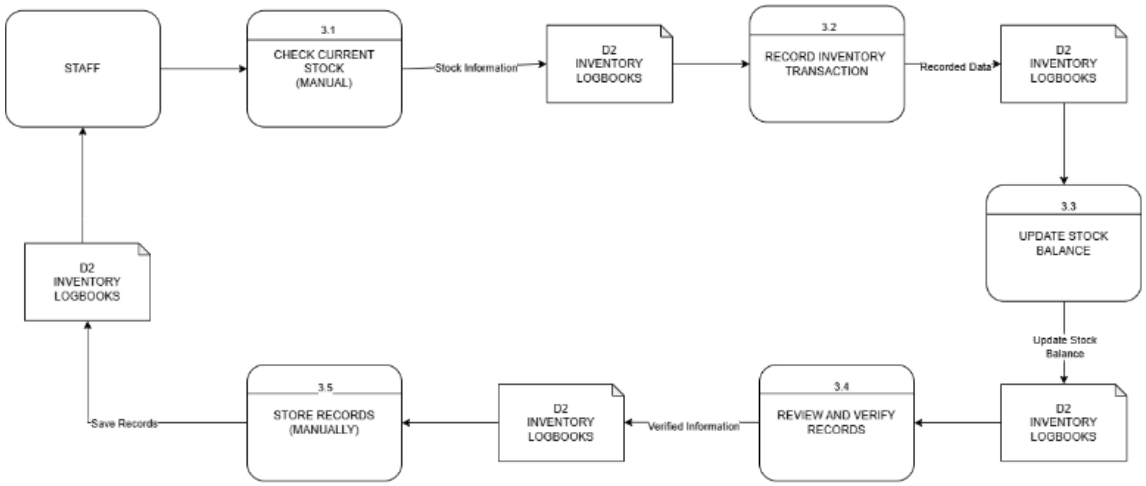


Figure 2.2 - This diagram provides a detailed view of the inventory recording process in the current system. Inventory transactions are manually documented using logbooks and paper-based records to track stock availability, usage, and replenishment.

Figure 2.3 Proposed Context Diagram

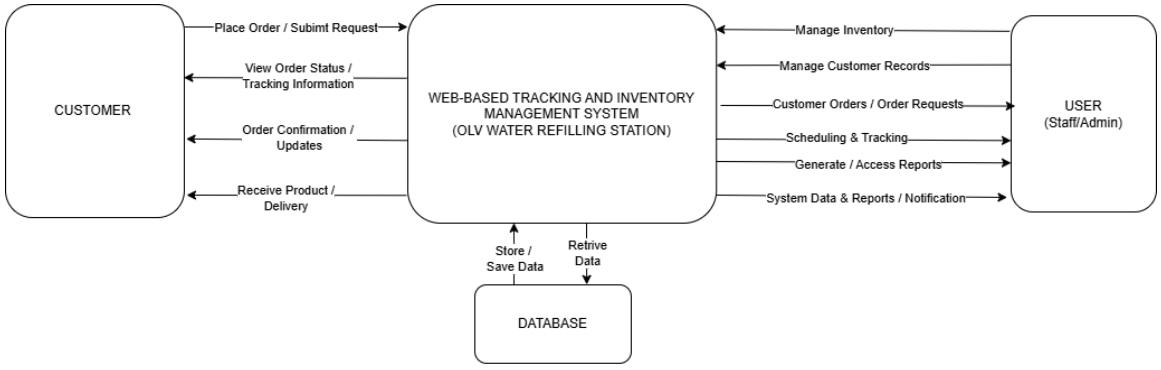


Figure 2.3 - This diagram shows the overall flow of information in the proposed Web-Based Inventory and Tracking Management System for OLV Water Refilling Station. It illustrates the interaction between system users and the centralized system in managing inventory, customer records, scheduling, tracking, and reporting functions

.

Figure 2.4 Proposed Level 1 Data Flow Diagram

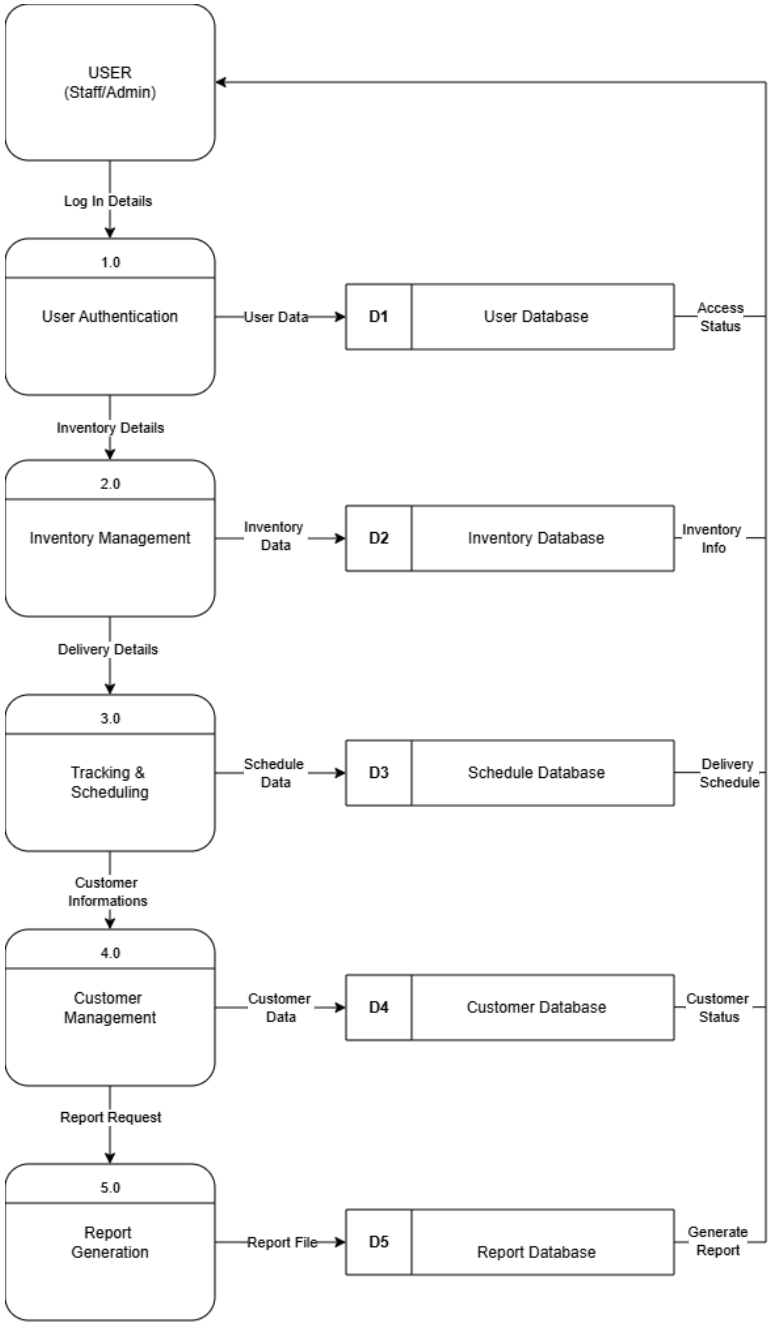


Figure 2.4 – This diagram presents the main modules of the proposed system, which include user authentication, inventory management, tracking, customer management, and

report generation. Each module is connected to a centralized database to ensure data consistency, accuracy, and efficient record management.

Figure 2.5 Proposed Level 2 Data Flow Diagram – Inventory Management

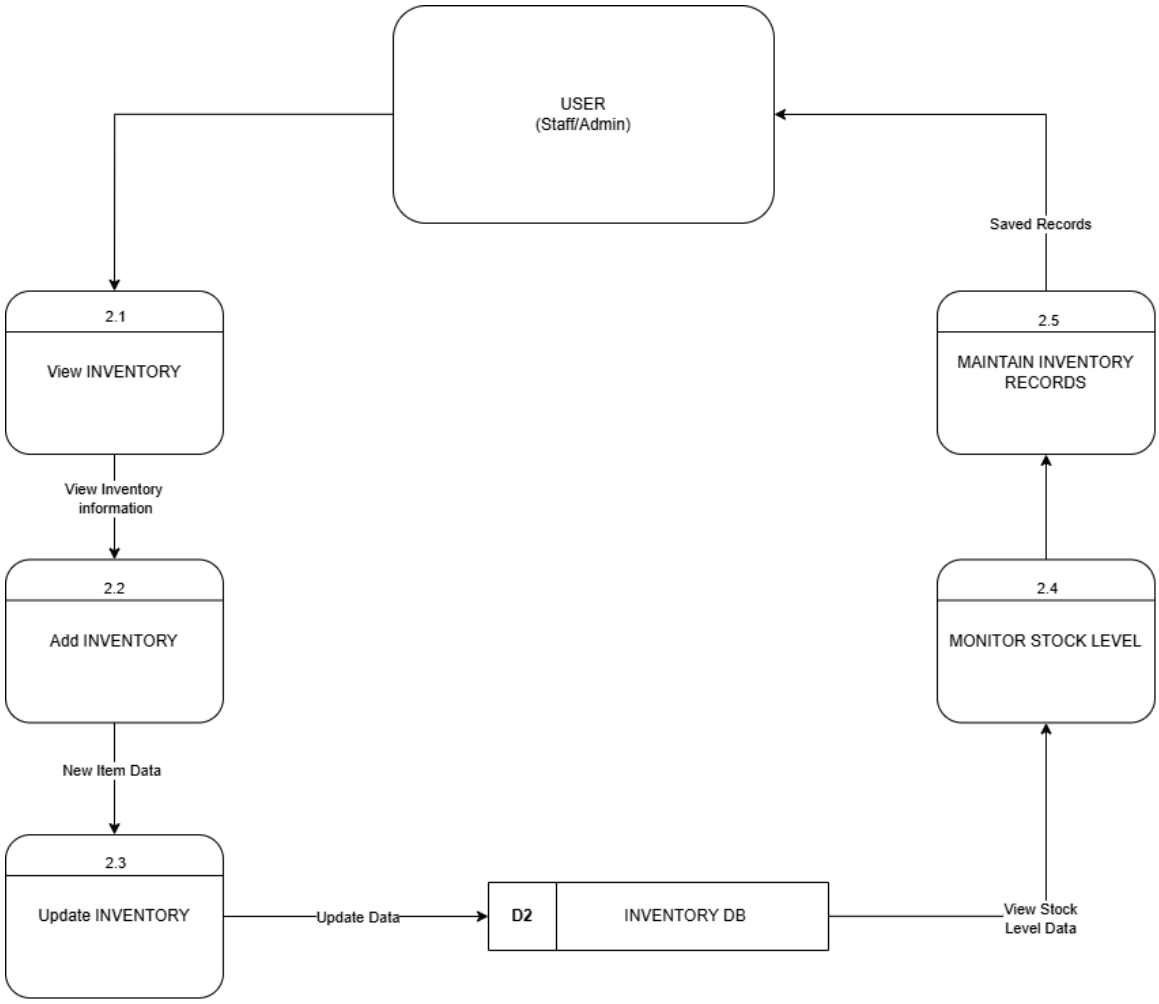


Figure 2.5 - This diagram illustrates the processes involved in managing inventory records in the proposed system. It includes adding, updating, viewing, monitoring, and maintaining stock information to ensure real-time and accurate inventory tracking.

Figure 2.6 Proposed Level 2 DFD – User Authentication

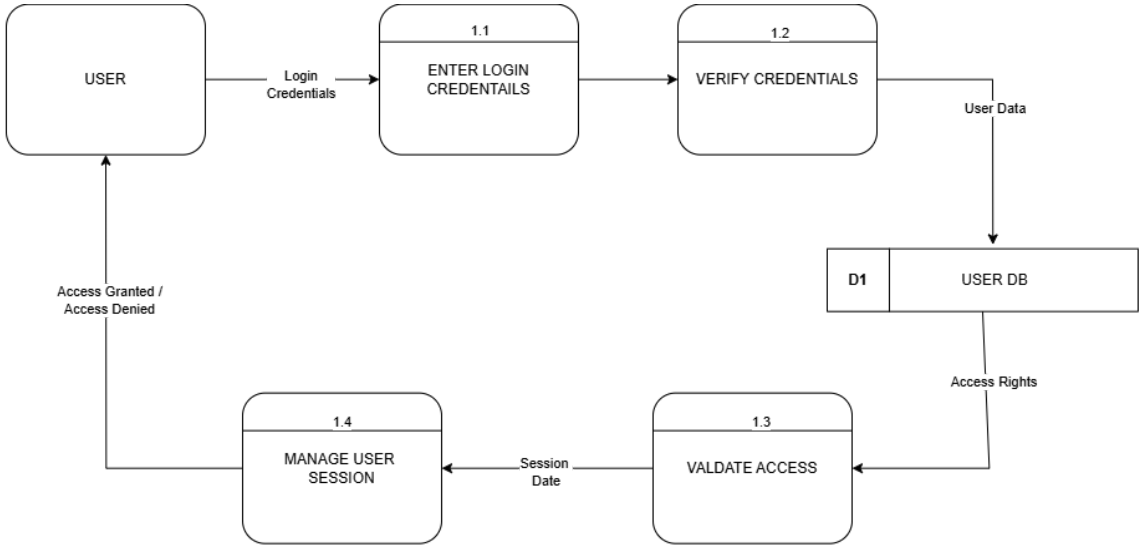


Figure 2.6. - This diagram shows the user authentication process, including login, credential verification, access validation, and user role management. These processes ensure system security and controlled access to system features.

Figure 2.7 Proposed Level 2 DFD – Customer Management

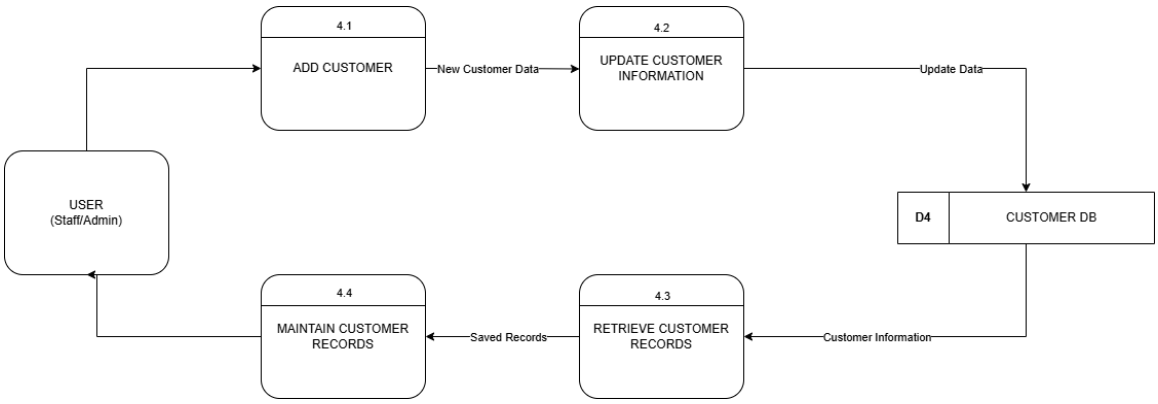


Figure 2.7 This diagram presents the processes involved in managing customer information. It includes recording, updating, retrieving, and maintaining customer data to support efficient transactions and improved customer service.

Figure 2.8 Proposed Level 2 DFD – Scheduling and Monitoring

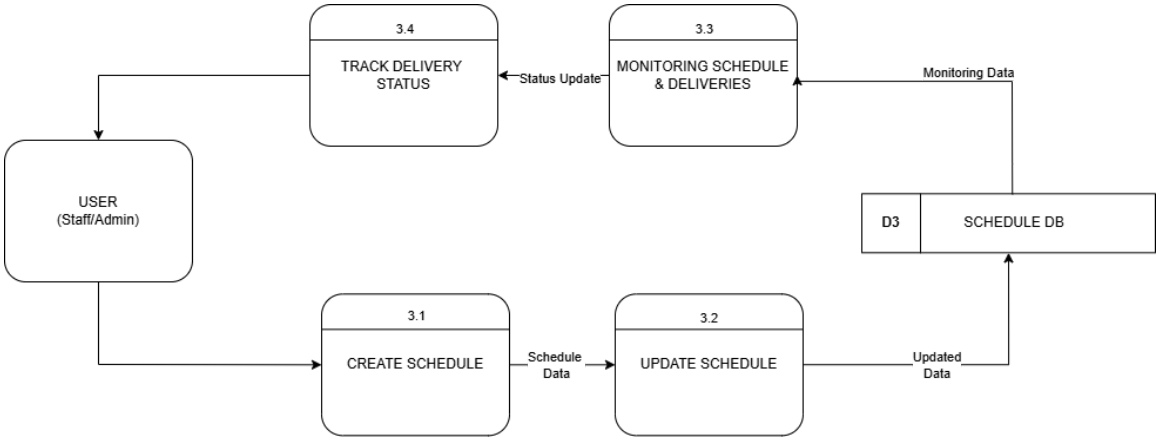


Figure 2.8. This diagram illustrates the processes used in creating, updating, and monitoring delivery schedules and operational activities. It ensures timely service delivery and effective tracking of customer orders and requests

Figure 2.9 Proposed Level 2 DFD – Report Generation

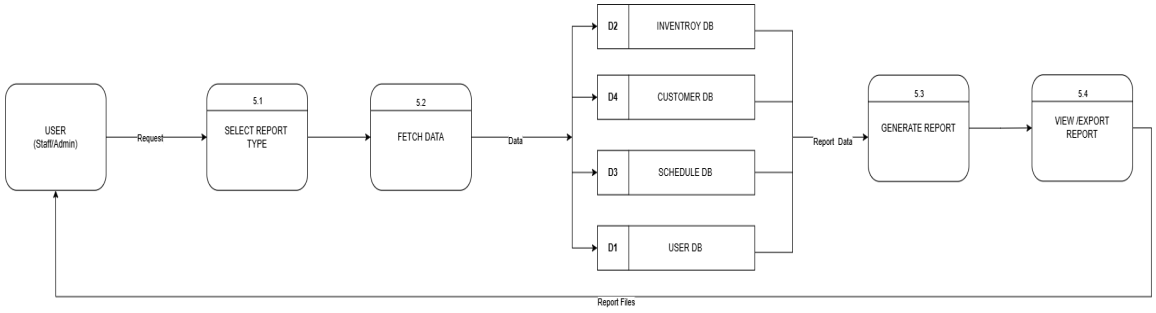


Figure 2.9 This diagram shows the process of generating system reports related to inventory, customers, schedules, and overall system activities. These reports provide accurate and organized data to support decision-making and operational analysis.

APPENDIX E. Transcript

TRANSCRIPT

Respondent:

Business Owner

Renato P. Lim

April 27, 2026 (First Interview on site)

Interviewer: Goodmorning ano po an pangaran kan saindong istasyon nin pagpuno nin tubig?

Client: OLV refilling station

Interviewer: gaano na po kahaloy ang saimong negosyo?

Client: 10 years na

Interviewer: sain po ining location nyo po?

Client: dgd ini sa 1095 St Agatha RD 5th st OLV Bitano Legazpi.City

Interviewer: Pirang empleyado an igwa ka ngunyan?

Client: duwa man lang ta minsan pag wara man ako gibo nagtatabang man ako para bako makunsumo sa oras

Interviewer: Anong mga produkto po asin serbisyo an saimong itinatao?

Client: huning purified water per bottle special occasions/orders

Interviewer: Ano an saimong mga oras nin operasyon?

Client: mga walong oras masarado kami mga 8pm

Interviewer: Paano mo po inaasikaso an mga order kan mga customer? (pag-agi, paghatod, pag-text, pagapod)

Client: Sa pag-text sainda kung kukuaon ninda an tubig o kung ipapadeliver na sana. Kung dae sinda agad nagreresponde sa text, tinatawagan pag-apod ko sinda. Nagagamit man ako nin Messenger bilang dagdag na paagi nin komunikasyon.

Interviewer: Paano binabayadan kan mga customer an saindang mga order?

Client: GCash Online na pagbayad An customer nagpapadara nin bayad sa paagi kan GCash

Pwedeng maglaog nin screenshot patunay nin pagbayad, Cash on Delivery Magbabayad ang customer gamit ang cash on delivery Karaniwan para sa mga walk-in at delivery orders

Interviewer: Paano mo narerecord an mga order asin benta kan mga customer?

Client: Sa logbook ko nirerekord an order asin benta para may maayos asin permanenteng talaan kan mga transaksyon asin madali ma-track an sales.

Interviewer: Paano mo po binabantayan an imbentaryo (mga lalagan nin tubig, mga suplay)?

Client: Kapag mataas an water bill, indikasyon ini na dakul an produksyon kaya mas mataas an sales. Samantalang kung mababa an water bill, mababa man an produksyon kaya mas mababa an sales.

Interviewer: Paano mo pinamamahalaan ang mga iskedyul ng paghahatid?

Client: Sa notebook ko inaayos an schedule nin delivery base sa oras nin order asin lokasyon kan Customer.

Interviewer: Paano mo sinusubaybayan ang impormasyon ng customer?

Client: Sa notebook or log book ko nirerekord an impormasyon kan mga customer—pangaran, address, asin contact number.

Interviewer: Magkano ang tubig mo kada galon?

Client: Mga 30 to 40 an kada galon kan tubig. Kung magpapadeliver, may dagdag na 5 delivery charge. Nagpaparenta man kami nin dispenser para sa mga okasyon arog kan kasal asin karinderya, kadalasan naaabot sa mga 30 gallons o mas dakul kada buwan depende sa gamit. Nagbebenta man kami nin sariling gallon sa halagang 170, asin kung magpapa-refill sinda, 30 kada galon kaya bale umaabot sa 200 an total kung may kasama nang gallon asin refill.

Interviewer: Pirang produkto nin tubig an saimong ipinapabakal sa sarong tipikal na aldaw?

Client: 100 gallons

Interviewer: Ano an mga problema na saimong naeeksperyensiyahan sa pag manage nin mga order asin delivery?

Client: Nagkakaigwa nin late delivery, may nalilito o nadodoble na orders, asin nahihirapan sa scheduling kan paghatod.

Interviewer: Nakakaeksperyensya ka daw nin kadepisilan sa pagsubaybay sa imbentaryo?

Client: Nahihirapan sa pag-track kan containers o gallons lalo na kung naipick-up na o idedeliver pa, kaya minsan dae klaro an status ninda.

Interviewer: Igwa daw nin mga pagkaatraso o mga sala sa pagrekord nin mga order?

Client: Wara man Maayos naman an pagrekord kan order kaya dae ako nakakaexperyensya nin sala o pagkaatraso

Interviewer: Ano an mga angat na inaatubang mo sa pamamahala kan mga record kan mga customer?

Client: Isa sa mga challenges ko sa pag-manage kan customer records iyo an pag-change kan cellphone number o contact number and yung fb acc ninda kan customer halimbawa ma chat ngaya mapa deliver o kaya aaramon mo kung sisay ini o iyo palan ini yung nag oorder sako . May mga pagkakataon na dae na updated an record kaya nagiging hirap kontakon an customer para sa delivery o follow-up.

Interviewer: Ano an mga pagpapauswag na gusto mong mahiling sa saimong presenteng sistema tanganing magin mas madali asin mas episyente an mga operasyon?

Client: Gusto ko yung mas simple asin mas organisado na sistema na makakapadali sa pag-manage asin pag-track kan mga order asin mas maaga o real-time na makita an mga order kan customers, para dae na manual asin maiwasan an kalituhan sa records.

Interviewer: Makakatabang daw an sistema na nagsusubaybay sa mga order asin delivery sa totoong oras? Nata?

Client: iyo syempre, makakatabang an real-time tracking kasi mas madali ma-monitor an order asin delivery asin maiwasan an delay.

Interviewer: Gurano kahalaga an pagsubaybay sa imbentaryo sa saimong aroaldaw na operasyon?

Client: Importante an inventory tracking para ma-monitor an suplay asin bilang kan gallons asin

maiwasan an kakulangan.

Interviewer: An mga automated notifications (SMS) para sa mga customer makakapakarhay daw kan saimong serbisyo?

Client: iyo imporatente iyan, lalo na para sa order confirmation at payment updates.

Interviewer: Ano an mga tampok na gusto mo sa sarong sistema nin pamamahala na nakabase sa web?

Client: Gusto ko nin sarong web-based system na simple asin parang Facebook page lang, kung sain madali makipag-communicate sa customer para sa orders asin inquiries. Mas gusto ko an may chat o messaging feature para mabilis an transaksyon asin komunikasyon.

Interviewer: Sa hiling mo igwa nin sumaryo sa dashboard (mga benta, mga order, estado kan

imbentaryo)? Nata?

Client: Oo, makakatabang an dashboard summary para mas madali ma-monitor an benta, order, asin imbentaryo.

Interviewer: muya mo po na gumamit nin sistemang nakabase sa web?

Client: iyo, muya ko gumamit nin sarong web-based na sistema kung makakapaayos asin makakapadali ini sa pag-manage kan negosyo.

Interviewer: Sa hiling mo madali para sa mga tawohan na matuto na makasunod?

Client: iyo madali lang matutunan kan mga staff an sistema ta simple man lang iyam asin user-friendly an interface.

Interviewer: Ano an saimong mga kahaditan manungod sa paggamit nin sarong sistema?

Client: Concern ko kung online o offline an system, basta stable asin dae nagkaka-issue sa gamit.

Interviewer: Ano ang magiging dahilan para maging user-friendly ang sistema para saimo?

Client: User-friendly an sistema kung simple, madali gamiton, asin dae komplikado.

Interviewer: Anong mga report an kaipuhan mo tanganing mabantayan an mga operasyon kan saimong negosyo, arog kan imbentaryo, paghatod, asin mga rekord kan mga customer?

Client: Gusto ko yung mga reports na makakapag-monitor kan imbentaryo arog kan containers, sales report, listahan kan mga utang o credit, delivery report, asin resibo o gas-receipt para sa mga transaksyon, para mas organisado asin madaling ma-track an operasyon kan negosyo.

Interviewer: Ano an mga komun na reklamo kan mga customer?

Client: Reklamo kan customers ko an late delivery asin late na messages o updates sa order.

Interviewer: Paano mapakarahay kan teknolohiya an saimong serbisyo?

Client: Makakapabuti an teknolohiya sa serbisyo sa pagin mas accurate, reliable, asin mas mabilis an operations.

Interviewer: Ano an mga pagpapakarhay o pagbabago na gusto mong ipaotob tanganing maging mas makaskas asin mas episyente an saimong mga operasyon?

Client: Depende sa system na gagagwin nyo, gusto ko yung mas mabilis na order processing, tracking, and less manual work para efficient an operations.

Interviewer: Salamat sa saimong oras asin sa paghiras nin impormasyon manungod sa saimong negosyo.

Client: Warang anuman. Salamat man sa pagbisita asin sa pag-interview. Sana makatabang an mga impormasyon na naibahagi ko sa pagdevelop kan iyong capstone project.

APPENDIX F. Documentation

Legazpi Campus

April 21, 2026

RENATO P. LIM
Store Owner
OLV Water Refiling Station
5th Street Our Lady's Village Brgy. Bitano, Legazpi, City

Dear Mr. Renato P. Lim,

We, the third-year students of the Bachelor of Science in Information Technology program currently enrolled in **IT Capstone Project 1**, respectfully request your permission to use **OLV Water Refiling Station** as our client for our study.

The purpose of this activity is to gather information through interviews, observations, and surveys with your staff and, if permitted, your customers. The data collected will help us identify existing business processes and challenges in laundry operations, which will serve as the basis for the development of our proposed capstone project system intended to benefit your business.

At this stage, we do not yet have a finalized system title or detailed design. All information gathered will be treated with utmost confidentiality and will be used strictly for academic purposes only.

Should you have any questions or require further clarification, please do not hesitate to contact us. We sincerely hope for your kind consideration and approval of this request.

Thank you very much for your time and consideration.

Respectfully yours,

Research Team Members:

Lim, Reinalyn R.

Gomez, Juan Rafael M.

Echague, Jim kirby

Clemente, John Manro B.

Recommending Approval:

JAIME PINEDA
Thesis Coordinator

Approved by:

MARIA DIEJHEY B. MARTINEZ, MIT
Academic Head

MAY ANNE V. REVALE
School Administrator





